



A Similarity-based Method for Saving Runs in Factorial Experiments

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Abstract:

This paper presents a method for saving experimental runs during the planning stage of factorial experiments when resources for conducting a complete experimental design are limited. In this context, the method investigates the number of coincidences between experimental runs as a measure of similarity. The real advancement of this approach lies in its emphasis on the dynamic interactions between the runs, rather than just focusing on the relationships among the factors represented as columns. Despite its simplicity, the method shows impressive performance in numerical evaluations.

Keywords: coincidence measure, experimental run, Hamming distance, negligible contrasts, similarity.

Mathematics Subject Classification (2010): 62K15.

1 Introduction

The prevalence of missing data in practical scenarios significantly complicates model identification and misleads the statistical inference, particularly in small sample experiments (see, for example, [Motavaze and Talebi \(2023\)](#)). In many cutting-edge industrial studies, limited resources often restrict the ability to conduct complete planned experimental designs. An illustrative case is presented by [Grima et al \(2019\)](#), who introduced a 2^3 experiment constrained to just seven runs due to these resource limitations. This situation renders the replication of missing runs impossible, creating a clear need for a robust solution. To address this

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challenge, an effective estimation method is essential for balancing the demands of costly experiments with the constraints of time, materials, and budget. Several researchers have tackled this problem by employing negligible contrasts to estimate missing observations (see, [Draper and Stoneman \(1964\)](#), [Box \(1990\)](#), and [Xampeny et al \(2017\)](#) amongst). Furthermore, [Xampeny et al \(2018\)](#) conducted a thorough analysis of which omitted runs can be estimated with desirable statistical properties. Building upon this, [Grima et al \(2019\)](#) proposed a valuable method that defines a range of potential values for each missing result, introducing a computationally intensive simulation approach to determine which contrasts might be deemed negligible. Although their findings offer significant advancements over earlier methodologies, they can also lead to computational complexity and time delays. In light of the principles underpinning the Design of Experiments (DoE), which focus on optimizing experimentation to reduce waste while maximizing the value of collected data, we recommend a straightforward and computationally efficient method for estimating responses or saving runs under resource constraints. This approach not only aligns with the fundamental goals of DoE but also positions researchers to achieve more accurate results without compromising on resource utilization.

The paper is organized as follows: Section 2 presents a motivating example that illustrates the problem discussed in the subsequent sections. Section 3 focuses on estimating missing observations when a negligible contrast is known. In Section 4, we introduce the similarity-based method and related concepts. Finally, Section 5 provides a numerical evaluation to highlight the results obtained from the proposed method, along with a discussion of the designed experiment used for the evaluation.

2 Motivation

Consider the 2^{7-4} fractional factorial design in [Box et al \(2005\)](#), to investigate how seven factors, labeled A to G, influence the time it takes for a cyclist to climb a hill. The details of the experimental design and the response variable are summarized in Table 1. These factors pertain to both the condition of the bicycle and the cyclist. However, let's assume that one of the experimental runs could not be conducted, either due to unforeseen circumstances or limited resources. For instance, we can consider a scenario where the cyclist is unable to perform the final experimental run because of time constraints.

Table 1: 2^{7-4} design and the response variable.

run	A	B	C	D=AB	E=AC	F=BC	G=ABC	y
1	-	-	-	+	+	+	-	69
2	+	-	-	-	-	+	+	52
3	-	+	-	-	+	-	+	60
4	+	+	-	+	-	-	-	83
5	-	-	+	+	-	-	+	71
6	+	-	+	-	+	-	-	50
7	-	+	+	-	-	+	-	59
8	+	+	+	+	+	+	+	88

3 Negligible contrasts

Estimating missing observations in a designed experiment requires the use of estimates from runs associated with negligible a priori contrasts. This means setting the algebraic expressions of the contrasts to zero, depending on the number of missing runs (see, [Box \(1990\)](#), [Xampeny et al \(2017\)](#), and [Grima et al \(2019\)](#)). By the hierarchy principle of factorial effects and considering the value of the ABC interaction to be negligible, we can write

$$-y_1 + y_2 + y_3 - y_4 + y_5 - y_6 - y_7 + y_8 = 0. \quad (3.1)$$

Using the Equation (3.1), we can estimate the missing run based on the observed values. For example, if y_8 represents the missing observation corresponding to the last run, we can deduce it as follows:

$$\hat{y}_8 = 69 - 52 - 60 + 83 - 71 + 50 + 59 = 78.$$

Let σ_y^2 is the variance of the response variable. From (3.1), $var(\hat{y}_8) = 7\sigma_y^2$, and for the main effect A we have

$$A = \frac{1}{4}(-y_1 + y_2 - y_3 + y_4 - y_5 + y_6 - y_7 + y_8),$$

and $var(A) = \frac{\sigma_y^2}{2}$. But, with an estimated run, for example $\hat{y}_8$, we have

$$A = \frac{1}{4}(-y_1 + y_2 - y_3 + y_4 - y_5 + y_6 - y_7 + (y_1 - y_2 - y_3 + y_4 - y_5 + y_6 + y_7)),$$

and $var(A) = \sigma_y^2$.

Although the method is straightforward, it significantly inflates the variance of estimated response variable, effectively doubling the variance of the factorial effects and complicating model identification. In the next section, we aim to address this issue and estimate the missing runs.

4 Similarity based method

In the experiment described in Section 2, which consists of 7 runs and has 6 degrees of freedom, we can estimate 6 factorial effects. Following the hierarchy of effects, we focus on the main effects $A - F$. Figure 1 illustrates the factorial plots for the effects we are considering.

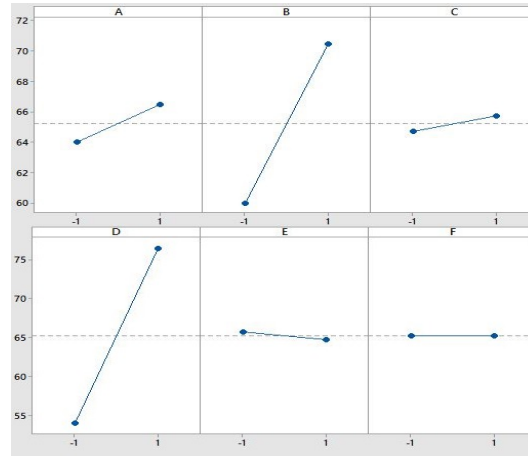


Figure 1: Factorial effects plot.

According to the results shown in Figure 1, we can exclude factors E and F from our analysis; and project the design onto factors A, B, C, and D, treating the data as an incomplete non-replicated factorial design, as illustrated in Table 2.

Table 2: The projected design and the response variable.

run	A	B	C	D	y
1	-	-	-	+	69
2	+	-	-	-	52
3	-	+	-	-	60
4	+	+	-	+	83
5	-	-	+	+	71
6	+	-	+	-	50
7	-	+	+	-	59
8	+	+	+	+	NA

4.1 Coincidence matrix

This section introduces a robust measure of coincidence designed to quantify the similarities between different runs; which allows for a nuanced understanding of how similar or dissimilar these runs are with respect to one another. The key innovation of this approach is its emphasis on investigating the relationships between the runs, which are represented as rows in the dataset, rather than merely focusing on the relationships between the factors, represented as columns. This shift in perspective can lead to more meaningful analyses and results.

Definition 4.1. For a design $D = [x_{ij}]_{N \times n}$ the number of coincidence between the i^{th} and j^{th} rows is

$$\delta_{ij}(D) = \sum_{k=1}^n \delta(x_{ik}, x_{jk}),$$

where $\delta(x, y)$ is the Kronecker delta function.

Remark 4.2. The quantity $H_{ij} = n - \delta_{ij}(D)$ is known as the Hamming distance between the i^{th} and j^{th} rows in algebraic coding theory.

Definition 4.3. Based on the Definition 4.1 for any design $D = [x_{ij}]_{N \times n}$ the coincidence matrix is defined as

$$M = \begin{pmatrix} n & \delta_{12} & \dots & \delta_{1n} \\ \delta_{21} & n & \dots & \delta_{2n} \\ \cdot & \cdot & n & \cdot \\ \delta_{N1} & 0 & \dots & n \end{pmatrix}$$

Remark 4.4. The coincidence matrix defined above, is a symmetric matrix.

Example 4.5. For the design in Table 2, the coincidence matrix is

$$M = \begin{pmatrix} 4 & 2 & 2 & 2 & 3 & 1 & 1 & 1 \\ 2 & 4 & 2 & 2 & 1 & 3 & 1 & 1 \\ 2 & 2 & 4 & 2 & 1 & 1 & 3 & 1 \\ 2 & 2 & 2 & 4 & 1 & 1 & 1 & 3 \\ 3 & 1 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 3 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 1 & 3 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 2 & 2 & 2 & 4 \end{pmatrix}$$

4.2 Grouping the runs

The coincidence matrix is utilized to group runs based on their similar distance values. Larger values indicate greater similarity between the runs. Therefore, the following grouping is recommended for the number of runs.

$$\mathcal{G}_1 = \{1, 5\}, \mathcal{G}_2 = \{2, 6\}, \mathcal{G}_3 = \{3, 7\}, \mathcal{G}_4 = \{4, 8\}. \quad (4.1)$$

Therefore, if y_8 is the missing run, we have $\hat{y}_8 = y_4$, and from the grouping in (4.1), $\text{var}(\hat{y}_8) = \sigma_y^2$. Therefore, the effects will have the same variance as if all runs had been conducted; and $\text{var}(A) = \frac{\sigma_y^2}{2}$.

5 Numerical evaluation

To effectively illustrate the advantages of the similarity-based method over the negligible contrast method, we conducted a numerical comparison of missing observations. In a randomized run order, any experimental run could end up being the last one, which might prevent the cyclist from completing it. Table 3 clearly presents the differences between our innovative approach and the traditional method that treats negligible contrasts as zero when estimating missing runs. This numerical analysis compellingly showcases the superiority of our method in effectively addressing the challenges posed by missing data, as well as, improvement in terms of variance estimates.

Table 3: The observations and their estimated values.

run	1	2	3	4	5	6	7	8
observed response variable	69	52	60	83	71	50	59	88
estimated value by negligible contrasts	79	42	50	93	61	60	69	78
estimated value by the proposed method	71	50	59	88	69	52	60	83

Discussion and Conclusion

This paper introduces an innovative method that leverages the relationships among runs (i.e., rows) instead of focusing solely on factors (i.e., columns). By identifying similar runs, we can effectively substitute their corresponding response values when faced with missing observations, ensuring more accurate and reliable outcomes.

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